

WEEK 4 – POWER BI EDA REPORT

SHOPIFY STOCK ANALYSIS USING POWER BI

1. INTRODUCTION:

In Week 4, I worked on Exploratory Data Analysis (EDA) using Microsoft Power BI. The dataset used for this activity contains historical stock information of Shopify.

The main aim of this project was to understand how Shopify stock prices and trading volume changed over different dates. Power BI was used to convert the raw stock data into meaningful charts, measures, filters, and an interactive dashboard.

During this activity, I worked with stock values such as Open, High, Low, Close, and Volume. I also created moving averages to understand short-term and long-term trends.

2. OBJECTIVES OF THE PROJECT:

The main objectives of this project were:

- To explore Shopify stock data.
- To understand changes in stock prices over time.
- To analyse daily trading volume.
- To create moving-average calculations using DAX.
- To compare short-term and long-term trends.
- To compare Open, High, Low, and Close prices.
- To apply date-based filtering.
- To create an interactive Power BI dashboard.

3. DATASET USED:

The Shopify stock dataset contains historical information recorded for different trading dates.

The important columns used in the analysis were:

Column	Description
Date	Date of stock trading
Open	Price at the beginning of the trading day
High	Highest price during the day
Low	Lowest price during the day
Close	Price at the end of the day
Volume	Number of shares traded

4. DATA PREPARATION:

Before creating the dashboard, the stock dataset was checked in Power BI.

The date column was connected with the Date Table so that time-based analysis could be performed. The numerical fields were also checked before creating the calculations.

5. AVERAGE TRADING VOLUME:

To understand the general trading activity, an average volume measure was created.

DAX Measure:

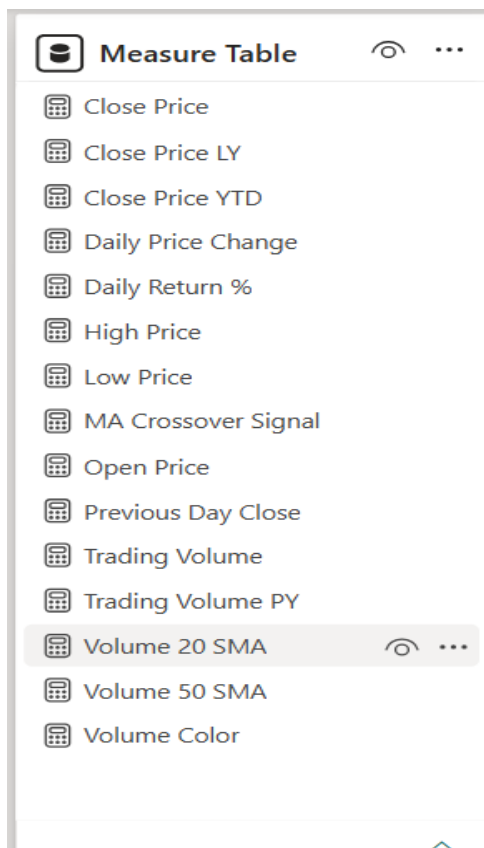
Average Volume =
AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume])

6. 20-DAY MOVING AVERAGE:

A 20-day moving average was created to study recent trading-volume behaviour.

DAX Measure:

```
Volume 20 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
RETURN  
CALCULATE(  
    AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
    DATESINPERIOD(  
        'Date Table'[Date],  
        CurrentDate,  
        -20,  
        DAY  
    )  
)
```

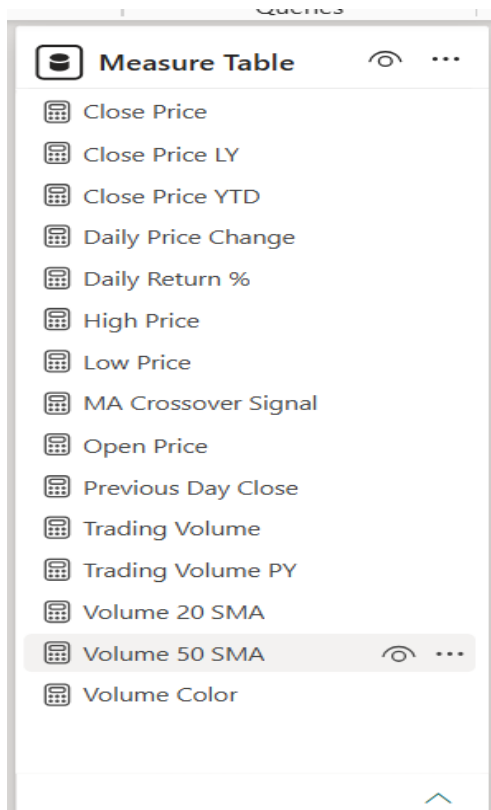


7. 50-DAY MOVING AVERAGE:

A 50-day moving average was created to provide a longer-term view of trading volume.

DAX Measure:

```
Volume 50 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
RETURN  
CALCULATE(  
    AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
    DATESINPERIOD(  
        'Date Table'[Date],  
        CurrentDate,  
        -50,  
        DAY  
    )  
)
```

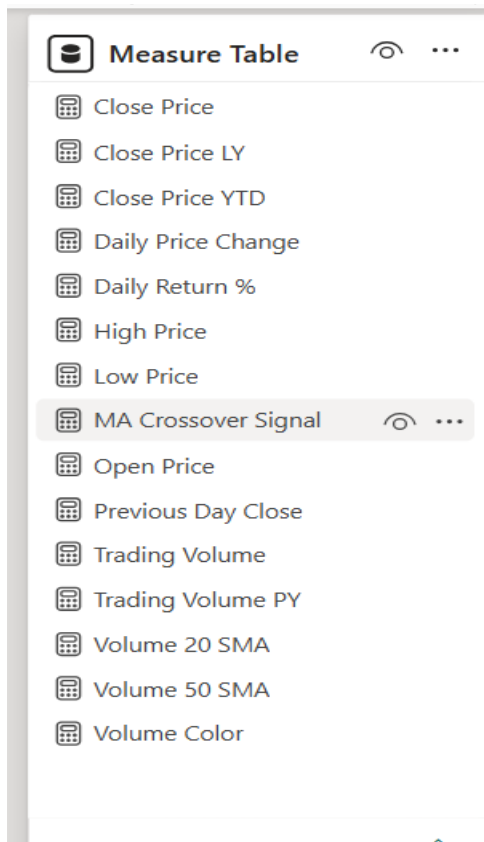


8. MOVING AVERAGE COMPARISON:

The 20-day and 50-day moving averages were compared to understand the short-term and longer-term trading patterns.

DAX Measure:

```
MA Crossover Signal =  
IF(  
    ISBLANK([Volume 20 SMA]) ||  
    ISBLANK([Volume 50 SMA]),  
    BLANK(),  
    IF(  
        [Volume 20 SMA] > [Volume 50 SMA],  
        "Bullish",  
        "Bearish"  
    )  
)
```



9. STOCK PRICE TREND:

A line chart was created to observe the closing price of Shopify stock over time.

Chart Configuration:

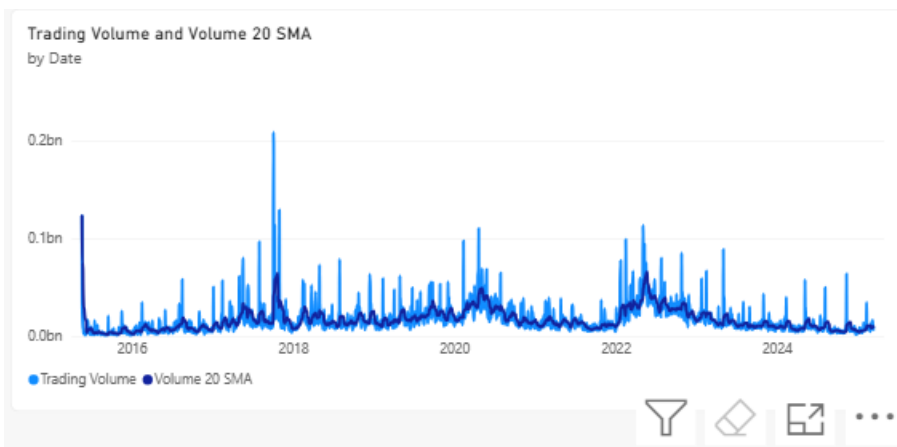
- X-Axis → Date
- Y-Axis → Close Price
- Visual → Line Chart

10. TRADING VOLUME ANALYSIS:

Trading volume was analysed using a separate visual.

The chart contains:

- Date
- Trading Volume
- 20-Day Average Volume

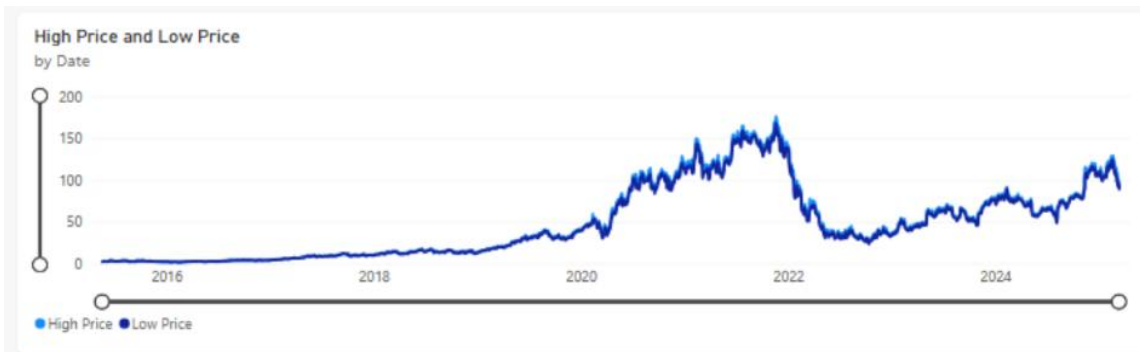


11. HIGH AND LOW PRICE COMPARISON:

A line chart was created to compare the highest and lowest stock prices.

Visual Configuration:

- X-Axis → Date
- Values → High Price, Low Price
- Chart → Line Chart

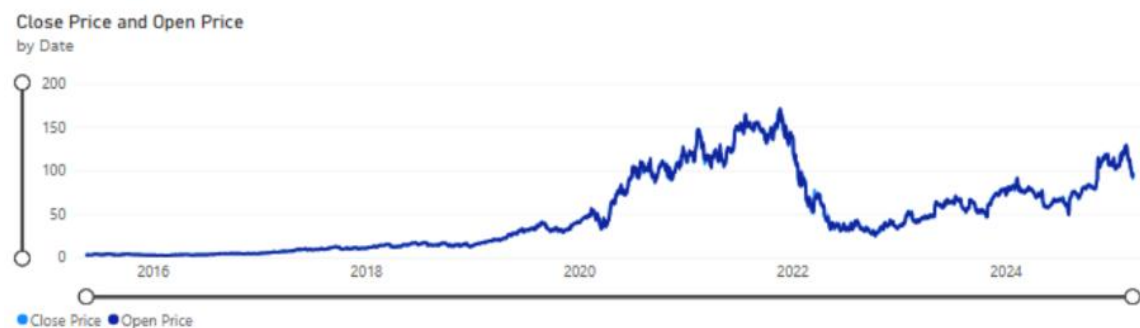


12. OPEN AND CLOSE PRICE COMPARISON:

Another line chart was created to compare Open Price and Close Price.

Visual Configuration:

- X-Axis → Date
- Values → Open Price, Close Price
- Chart → Line Chart

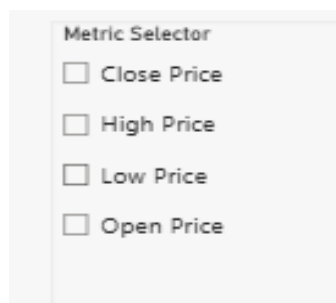


13. Metric Selector :

A Metric Selector was created using a Field Parameter. The selector includes four different price measures:

- ❖ Close Price
- ❖ High Price
- ❖ Low Price
- ❖ Open Price

The field parameter was created through: Modeling → New Parameter → Fields
This allows the user to switch between different stock price measures interactively without creating separate dashboards for each measure.

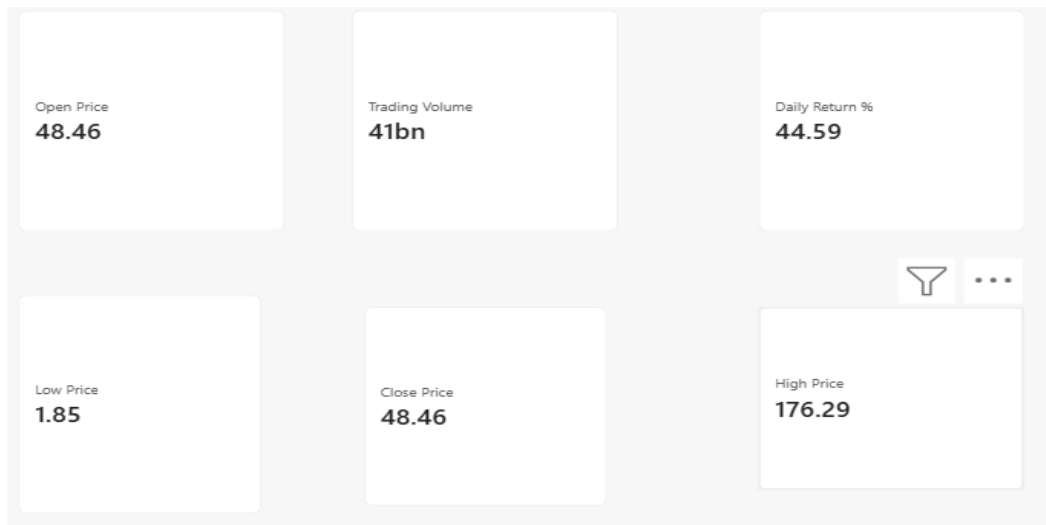


14. CUSTOM TOOLTIP:

A separate tooltip page was created to display additional stock information when the user moves over a chart point.

The tooltip contains:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Daily Change

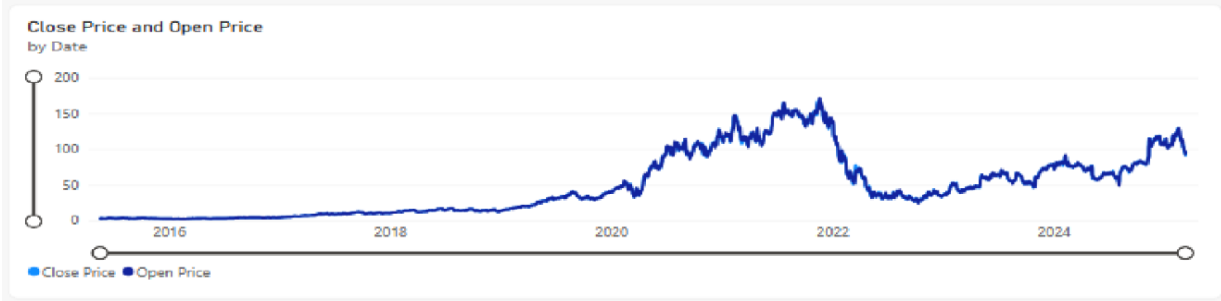
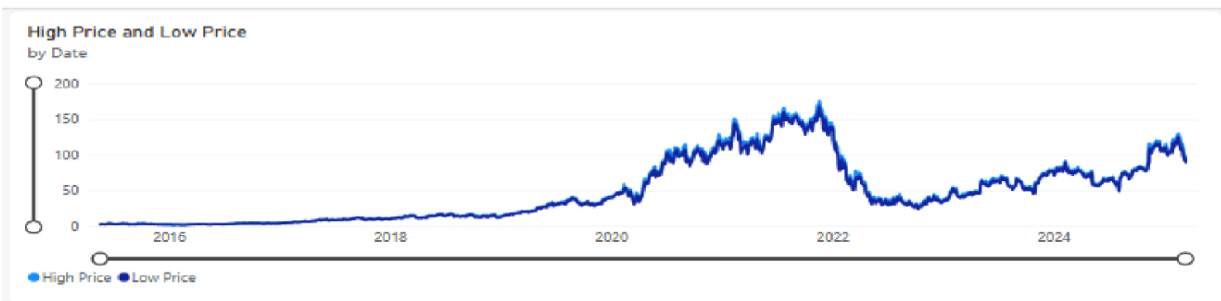
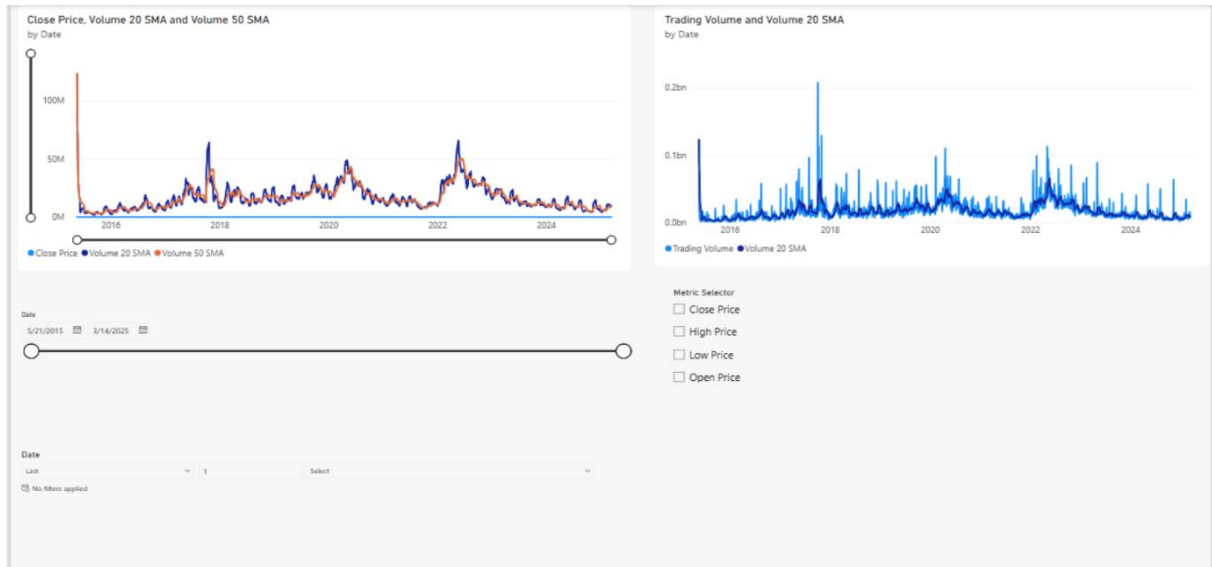


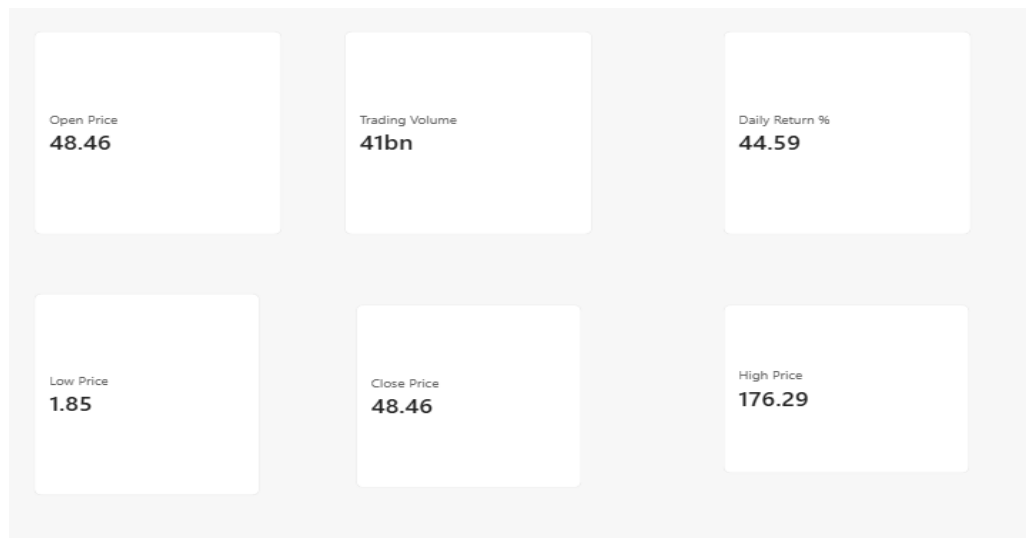
15. FINAL DASHBOARD:

All the created charts, measures, filters, and interactive features were arranged together to form the final Power BI dashboard.

The dashboard contains:

1. Stock price trend
2. Trading volume
3. 20-day moving average
4. 50-day moving average
5. High and Low price comparison
6. Open and Close price comparison
7. Date slicer
8. Relative date filter
9. Price selector
10. Custom tooltip





16. KEY OBSERVATIONS:

The analysis helped me understand the following:

- Shopify stock prices changed across different dates.
- Open and Close prices showed daily price movement.
- High and Low prices showed the daily price range.
- Trading volume changed depending on the trading period.
- The 20-day SMA showed the recent trading-volume pattern.
- The 50-day SMA provided a longer-term view.
- Comparing the two moving averages helped identify their relationship.
- Date filters made period-wise analysis easier.
- The Field Parameter made the dashboard more interactive.
- The tooltip provided additional information without creating extra charts.

17. CONCLUSION:

The Week 4 activity focused on analysing Shopify stock data using Microsoft Power BI. The project included stock-price analysis, trading-volume analysis, moving averages, date filtering, dynamic price selection, and custom tooltips.

The different visuals helped me understand stock behaviour from multiple perspectives. The DAX measures also gave me practical experience in performing calculations on time-based data.

Overall, this activity helped me improve my skills in Power BI, DAX, data visualization, and Exploratory Data Analysis.

Submitted by:

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